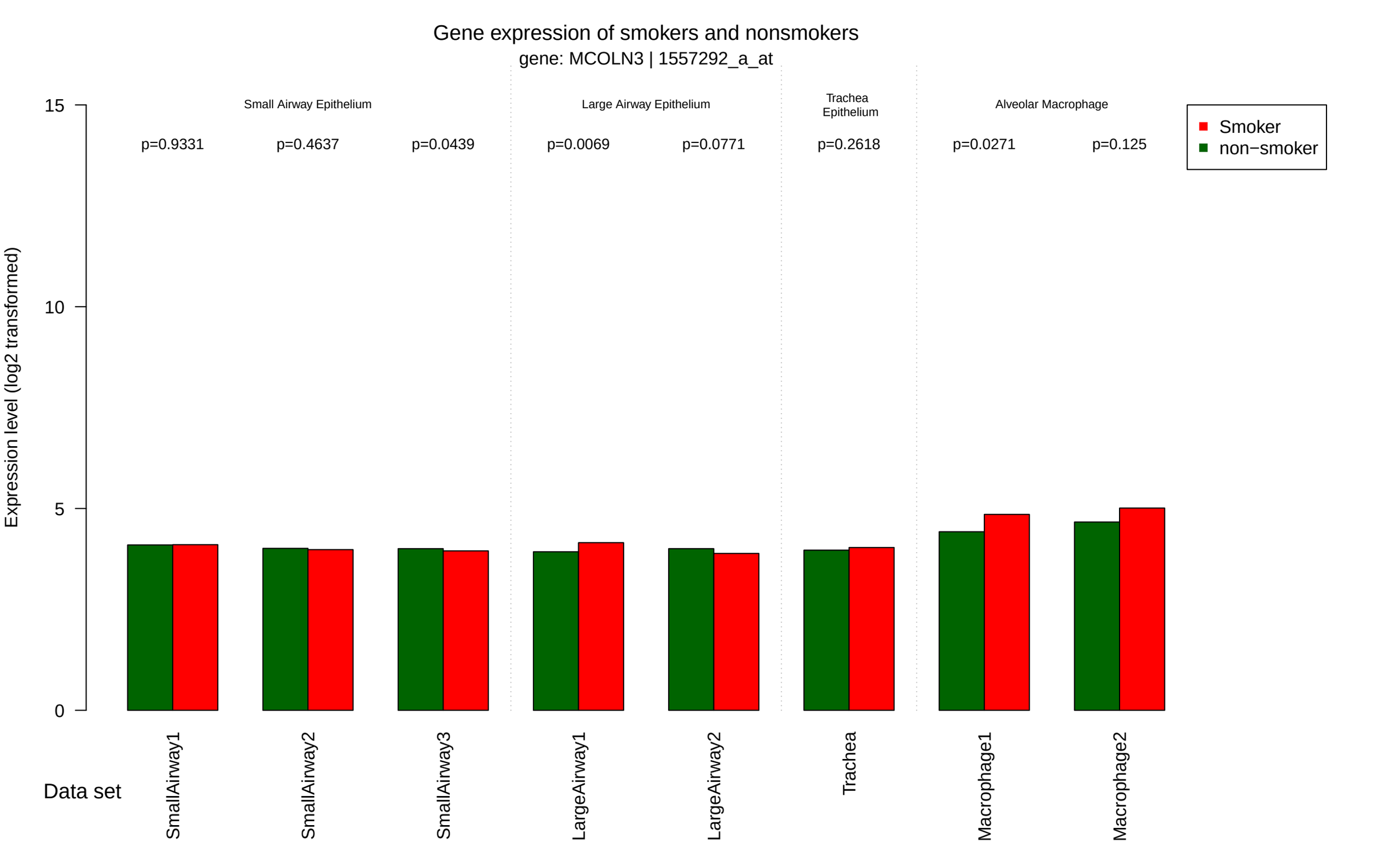
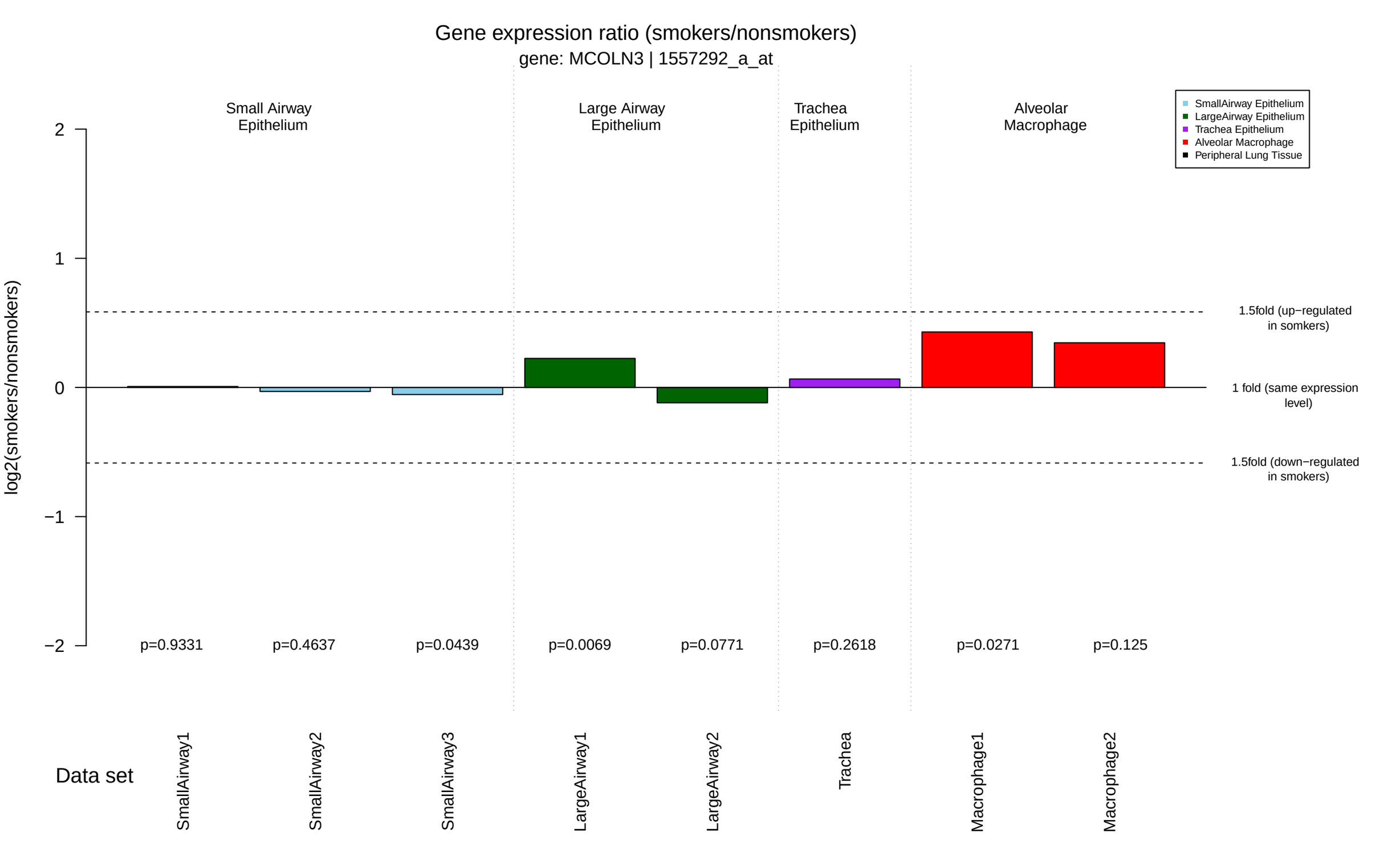
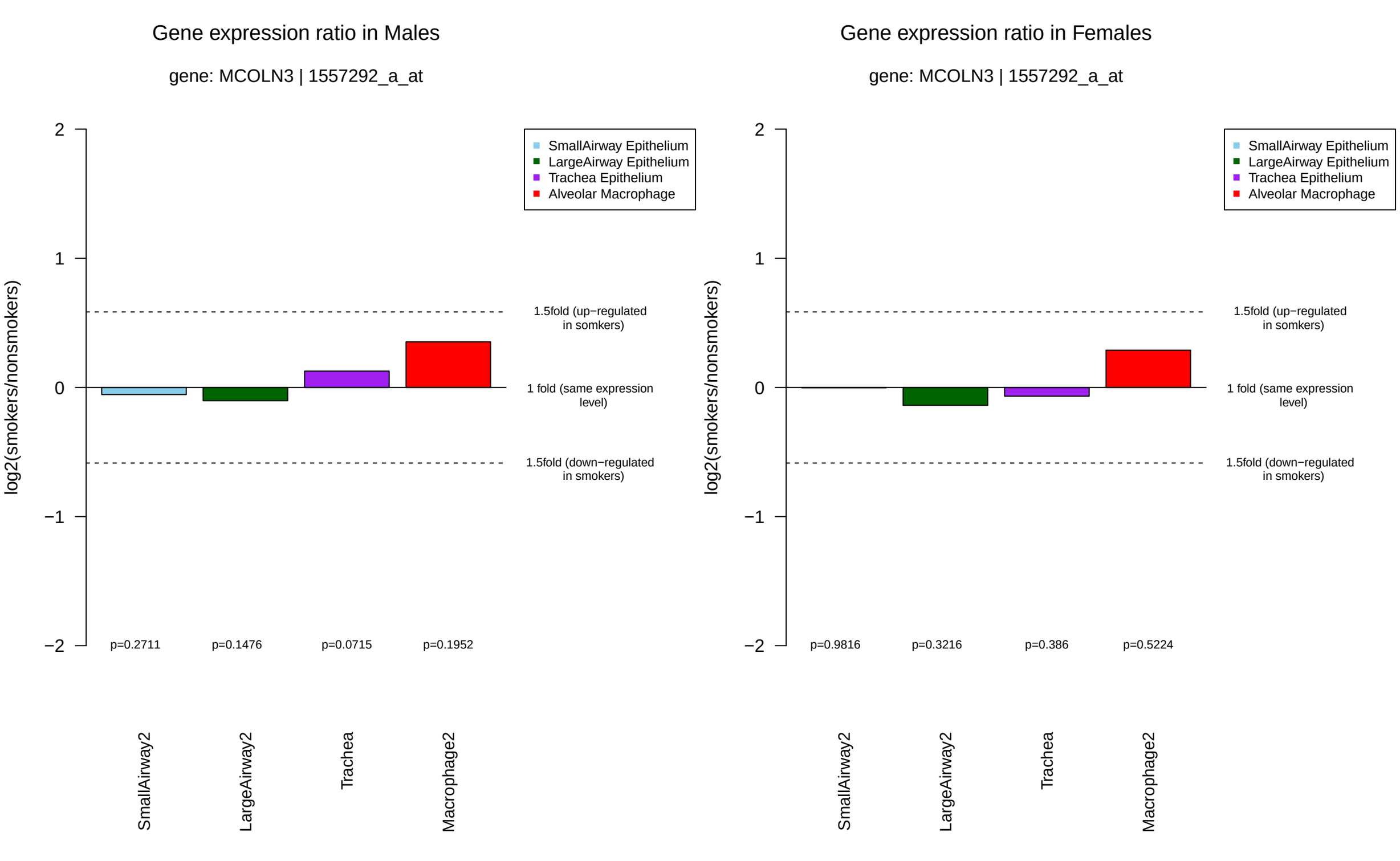
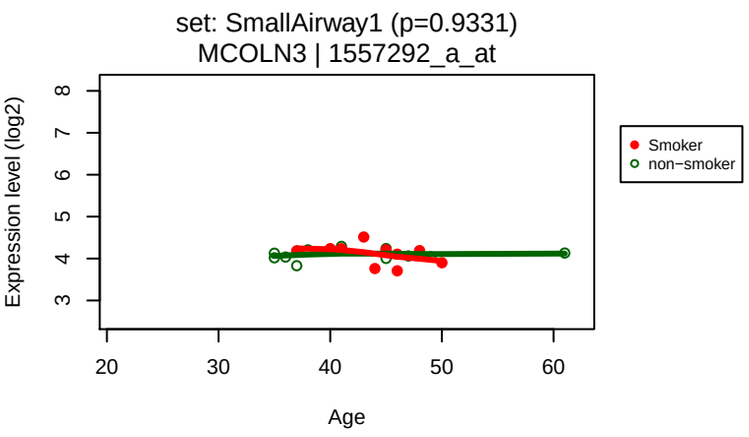
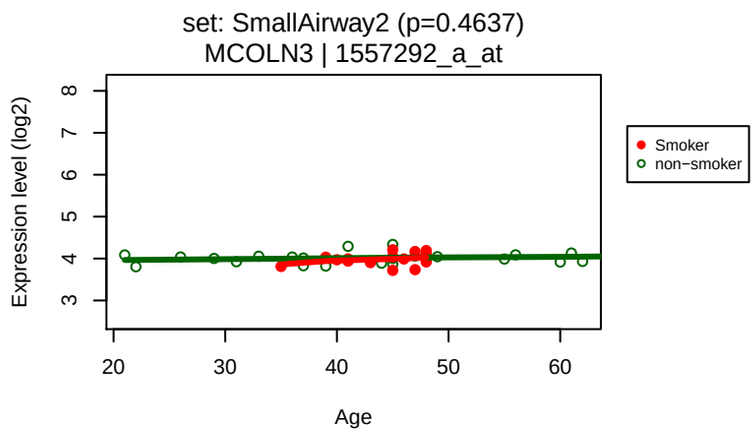
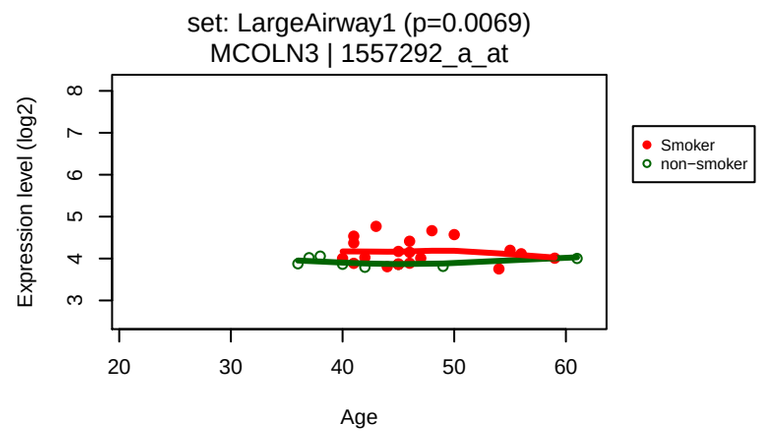

Gene expression of smokers and nonsmokers
gene: MCOLN3 | 1557292_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9331
p=0.4637
p=0.0439
p=0.0069
p=0.0771
p=0.2618
p=0.0271
p=0.125
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

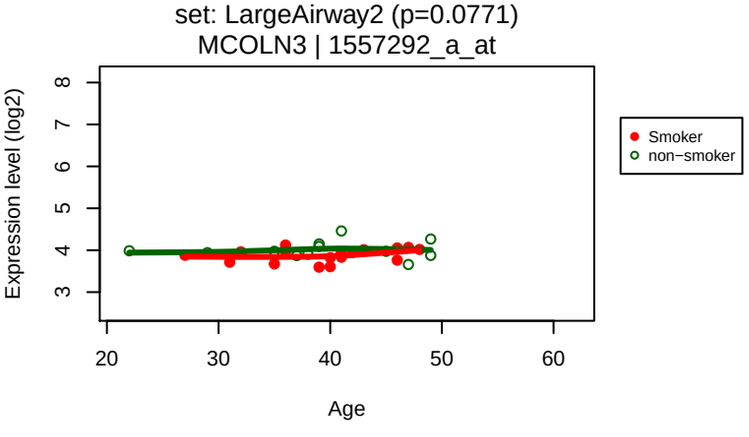

Gene expression ratio (smokers/nonsmokers)
gene: MCOLN3 | 1557292_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9331
p=0.4637
p=0.0439
p=0.0069
p=0.0771
p=0.2618
p=0.0271
p=0.125
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

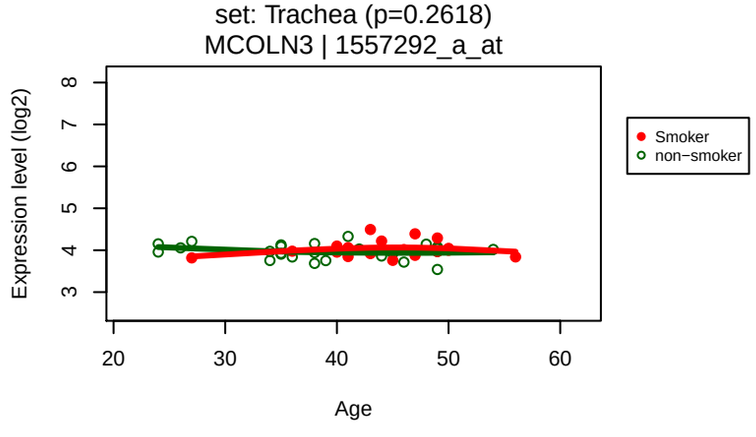

Gene expression ratio in Males
gene: MCOLN3 | 1557292_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2711
p=0.1476
p=0.0715
p=0.1952
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MCOLN3 | 1557292_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9816
p=0.3216
p=0.386
p=0.5224
SmallAirway2
LargeAirway2
Trachea
Macrophage2

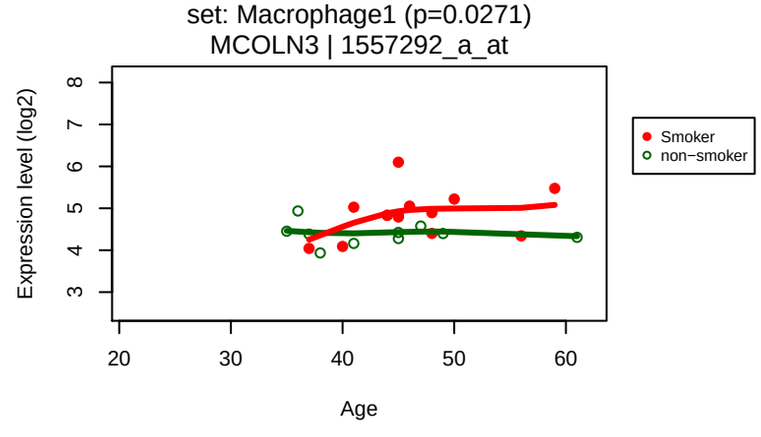

set: SmallAirway1 (p=0.9331)
MCOLN3 | 1557292_a_at
Expression level (log2)
Age
Smoker
non−smoker

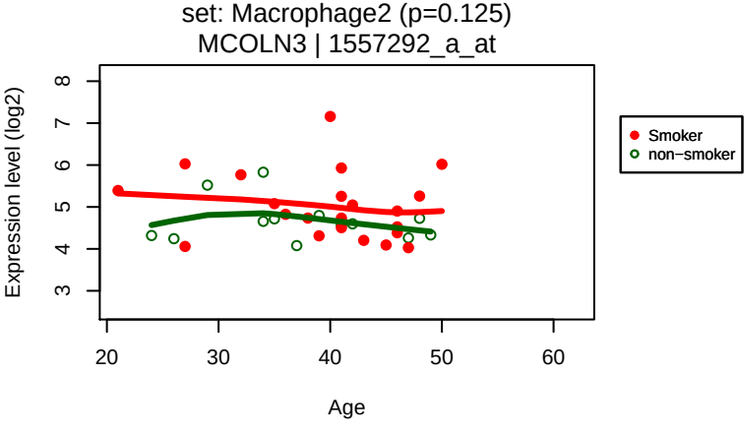

set: SmallAirway2 (p=0.4637)
MCOLN3 | 1557292_a_at
Expression level (log2)
Age
Smoker
non−smoker

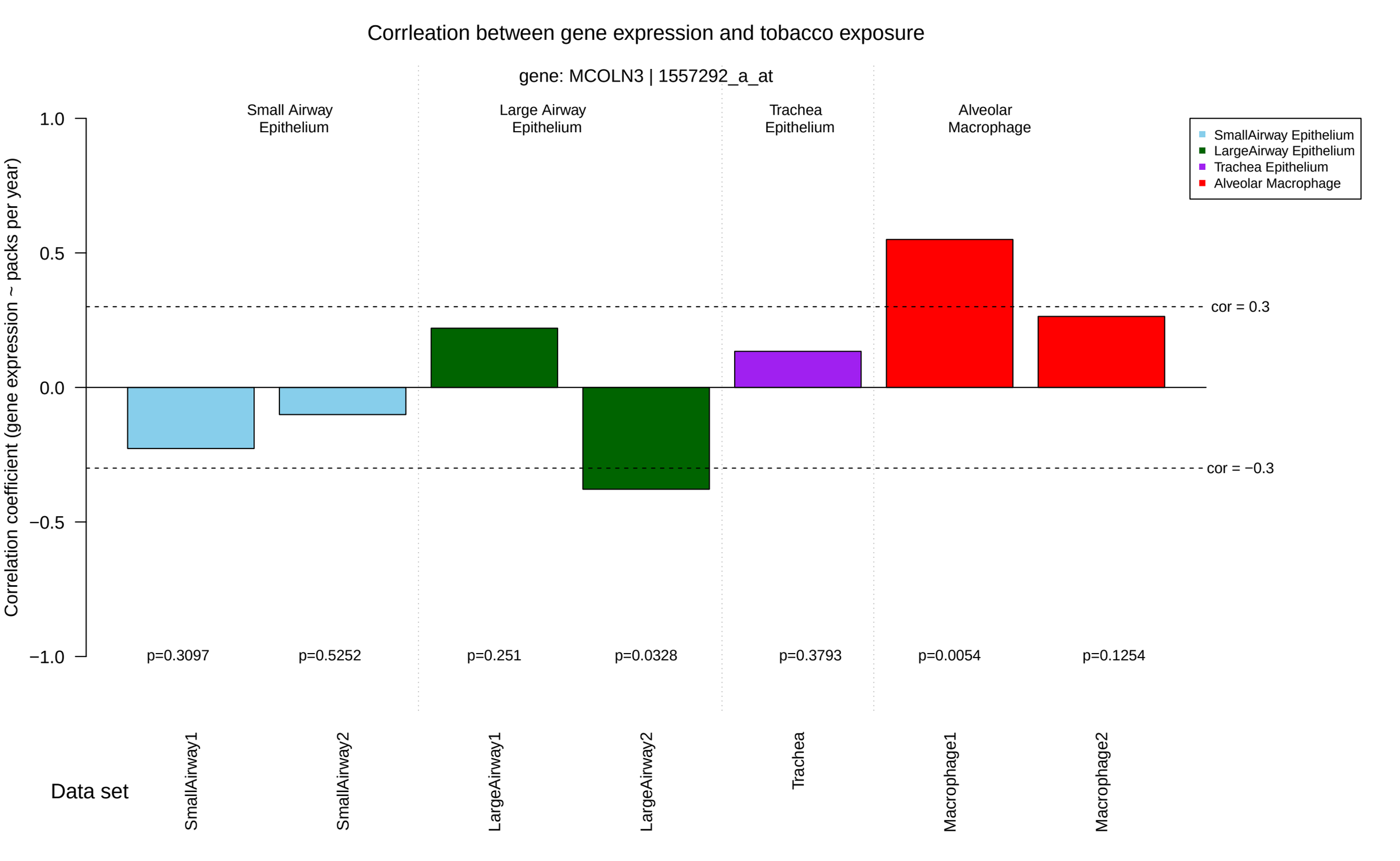

set: LargeAirway1 (p=0.0069)
MCOLN3 | 1557292_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0771)
MCOLN3 | 1557292_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2618)
MCOLN3 | 1557292_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0271)
MCOLN3 | 1557292_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.125)
MCOLN3 | 1557292_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MCOLN3 | 1557292_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3097
p=0.5252
p=0.251
p=0.0328
p=0.3793
p=0.0054
p=0.1254
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2